

# Keigo Kusumegi

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**Computational Social Scientist** and PhD candidate in Information Science with expertise in network science, machine learning, and data-driven methods to study innovation. Seeking to apply advanced analytical techniques to projects focused on **the broad impact of AI on societal impact**.

## Education

|                                                                                     |                       |
|-------------------------------------------------------------------------------------|-----------------------|
| <b>Cornell University</b> (Ithaca, NY), PhD program in Information Science          | Aug. 2023 –           |
| Advisor: Yian Yin                                                                   |                       |
| <b>University of Tsukuba</b> (Ibaraki, Japan), MSs in Policy and Planning Sciences  | Apl. 2021 – Mar. 2023 |
| <b>University of Tsukuba</b> (Ibaraki, Japan), B.S. in Policy and Planning Sciences | Apr. 2017 – Mar. 2021 |

## Publications

1. **Kusumegi, K.**, <sup>¶</sup> Yang, X., <sup>¶</sup> Ginsparg, P., De Vaan, M., Stuart, T., Yin, Y. Scientific production in the era of large language models. *Science*, 390, 1240-1243 (2025) [link](#)
2. **Kusumegi, K.**, <sup>¶</sup> Yang, X., <sup>¶</sup> Ginsparg, P., De Vaan, M., Stuart, T., Yin, Y. A robust association between LLM use and scientific productivity: Assessing stopping-time selection. *arXiv*, (2026) [link](#)
3. **Kusumegi, K.**, Yin, Y. The Specialization of interdisciplinary innovators in science and technology. (working paper)
4. **Kusumegi, K.**, Daniel E. A., Yukie S. Dissecting the Gender Divide: Authorship and Acknowledgment in Scientific Publications. *Scientometrics*, (2026) [link](#)
5. **Kusumegi, K.**, Sano, Y. Dataset of identified scholars mentioned in acknowledgement statements. *Scientific Data* 9, 461 (2022) [link](#)
6. **Kusumegi K.**, Sano, Y. 2021. Citations and Gender Diversity in Reciprocal Acknowledgement Networks. Preprint at *arXiv:2104.01729* (2021) [link](#)

<sup>¶</sup> denotes equal contributions

## Recent Conferences/Talks

- KDD 26
- KDD 25 SciSoc LLM Workshop
- Complex Networks 2025
- ICSSI 2025
- IC2S2 2025

## Research Experiences

|                                        |                     |
|----------------------------------------|---------------------|
| <b>Yin's Lab at Cornell University</b> | Aug. 2023 – Current |
| <i>Graduate Research Assistant</i>     | <i>Ithaca, NY</i>   |

- Quantified the influence of Large Language Models on scientific production using large-scale publication metadata analysis applying difference-in-differences frameworks
- Developed a systematic approach to estimate AI usage in scientific articles and quantify its impact on writing styles and citation behavior
- Evaluated the application of AI across various scientific fields to acceleration in scientific discovery
- Explored the complex interaction between individual knowledge expertise and project alignment

## Mentorships

- Supervised undergrad and master's student research projects, providing weekly guidance on methodology
- Provided technical assistance of coding and text analysis for Humanity PhD students at Summer Graduate Fellowship in Digital Humanities

### **Sano Lab at University of Tsukuba**

Apr. 2021 – Jun. 2023

#### *Graduate Research*

*Ibaraki, Japan*

- Uncovered hidden academic collaboration networks by parsing contribution acknowledgments in scientific articles, providing a deeper understanding of team dynamics

## **Research Internships and Industrial Experiences**

### **G-RIPS**

Jun. 2021 – Aug. 2021

#### *Research internship*

*Sendai, Japan*

- Designed and implemented an optimization algorithm of wireless base station placement using topology and graph theory

### **OPTiM Corporation**

Sep. 2020 – Oct. 2020

#### *Research internship*

*Tokyo, Japan*

- Developed CNN-based multi-object detection model for industrial quality control

### **MARK Creative**

Jan. 2019 – Jul. 2023

#### *Software engineer*

*Ibaraki, Japan*

- Designed and implemented graph database and its API for data analysis and system optimizations

## **Awards**

- |                                                                   |      |
|-------------------------------------------------------------------|------|
| • Funai Overseas Scholarship (\$120K)                             | 2023 |
| • QUAD Fellowship (\$50K)                                         | 2023 |
| • University of Tsukuba President's Commendation                  | 2023 |
| • Tsukuba Graduate Study Scholarship                              | 2023 |
| • Nakagawa Ikueikai Scholarship                                   | 2021 |
| • Best Poster Award                                               | 2021 |
| • Scholarship from the Japan Society for The Promotion of Science | 2018 |